

SMART INDIA HACKATHON 2026



- Problem Statement ID – SIH26055
- Problem Statement Title – Smart Scan strategy for Electronic Warfare
- Theme – Robotics and Drones
- PS Category – Software
- Team ID – 120057
- Team Name – Evinco

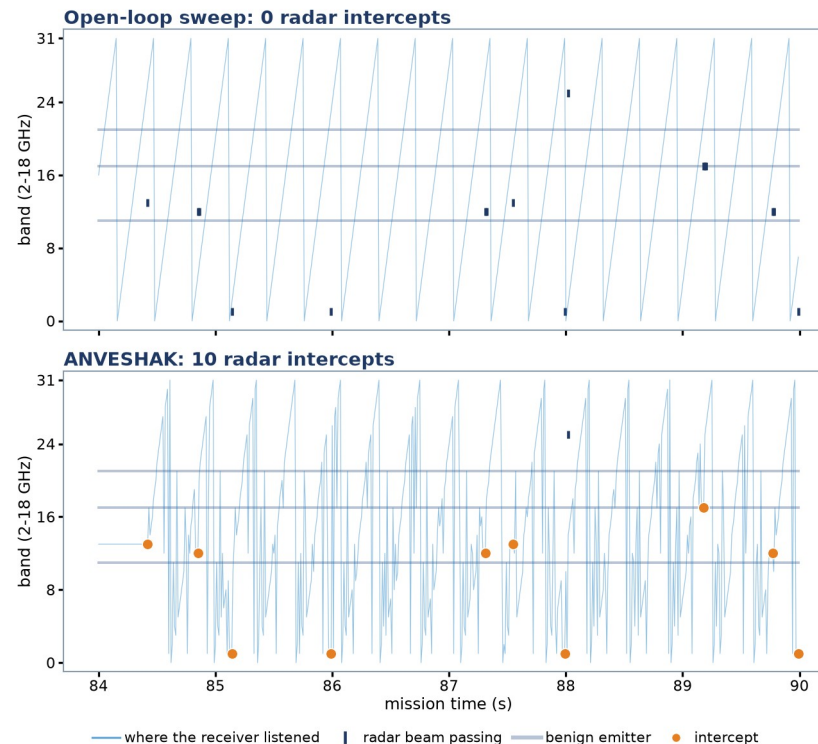


Proposed Solution :

ANVESHAK — a machine-learning scheduler for an Electronic Support receiver that learns, from its own hits and misses, where and when to listen. With no emitter library or pre-mission data, it discovers emitters, measures their scan periods and dwells exactly when their beams return, while still guaranteeing every band is searched.

Addressing the Problem :

- An open-loop sweep gives equal time to empty bands, harmless beacons and live threats.
- A fixed sweep can fall into lock-step with a radar's scan: a 32-band sweep never sees a 3.2 s radar for 15 of 16 phase offsets.
- Without prior intelligence, nothing tells the receiver where a new or threatening emitter is.



Innovation and uniqueness :

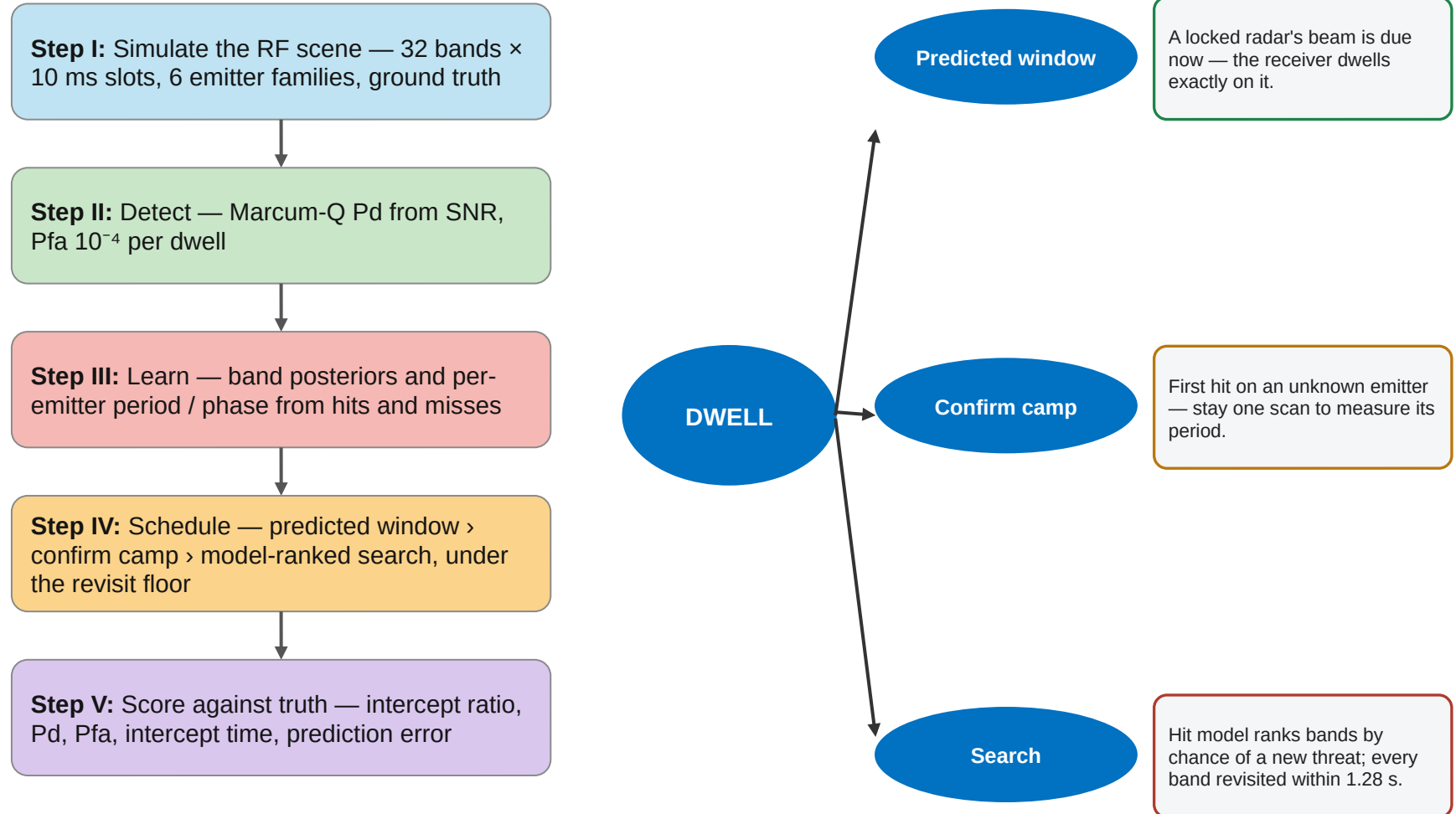
- Hit → confirm → lock: after a first hit it camps one scan to measure the period, then dwells only on predicted beam passes (95% within ± 20 ms).
- Learns from hits and misses: a gradient-boosted hit model, trained on simulator truth, ranks where to search — 15× better than chance at spotting a new threat.
- Earliest-deadline-first revisit floor: every band re-checked within 1.28 s — no learned policy can starve part of the spectrum.

Technologies :

Python 3.12 · NumPy · SciPy

- Marcum-Q detection model (Pd from SNR at set Pfa)
- Discounted Beta-Bernoulli Thompson sampling
- Approximate-GCD scan period / phase estimator
- XGBoost hit model trained on simulator truth
- Seeded RF scenario generator with ground truth
- FastAPI band-state console: CRUD + live scoring
- Docker, deployable on Microsoft Azure

TECHNICAL APPROACH



Feasibility :

Already running – the simulation core runs a 120 s, 32-band mission in about a second on one CPU core.

No hardware needed – a seeded RF environment with per-slot ground truth, exactly as the PS asks.

Low cost – NumPy / SciPy on commodity CPUs; no GPU for training or inference.

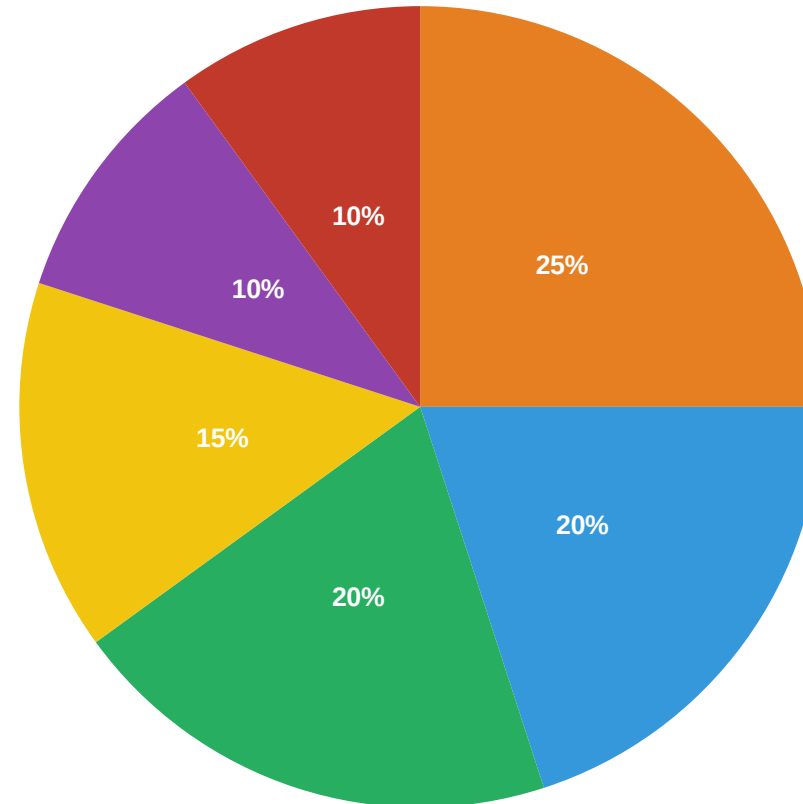
Modular design – environment, receiver, belief and scheduler swap independently; a real ES receiver plugs in behind the same interface.

Challenges in blind spectrum search (team estimate) :

No prior intelligence
No library, no frequency plan: every emitter is discovered, timed and tracked online from hits and misses.

Frequency-agile emitters
No stable period to learn; the hit model ranks their bands instead, our hardest case: 0.16 vs 0.12 for rules alone.

Pop-up threats
The revisit floor bounds blind time to 1.28 s per band, whatever the policy prefers.



Scan-on-scan sync
Commensurate periods hide radars from a fixed sweep; randomised search plus phase lock removes the trap.

Benign emitters
Only new intercepts earn reward, so always-on beacons are seen once and then left alone.

False period locks
Largest-consistent-period estimator; three missed predicted windows drop the lock.

Impacts :

- 4.8× the threat intercepts of an open-loop sweep
- 9× the sweep where radar scans fall in lock-step with it
- Next radar beam pass predicted within ± 20 ms, 95% of the time
- Hit model: 15× better than chance at finding new threats
- No band left unwatched for more than 1.28 s

4.8×threat-weighted
intercepts**9×**

in the sync trap

95%beam passes within ± 20
ms**15×**hit model vs chance
(AUC-PR)

Revisit guarantee: earliest-deadline-first floor $\rightarrow$ rules + hit-model scores $\rightarrow$ dwell.
Worst revisit in every simulated mission = 128 slots (1.28 s), exactly the bound.

Benefits :

Operational

- Threats are intercepted far more often, so warning and jamming cues are more reliable.
- Receiver time goes to new and threatening emitters, not known beacons.

Strategic

- Works with no pre-mission intelligence — suited to new theatres and unknown emitters.
- Measured scan periods and phases build an emitter library as a by-product.

Economic

- A software-only upgrade to existing ES receivers; no new RF hardware.
- The same receiver sensitivity yields more intercepts, extending the value of fielded sets.

A. Research

- Scan-on-scan: with a radar window of τ_r slots every T_r and a receiver window of τ_s every T_s , an intercept is possible for only $\min(1, (\tau_r + \tau_s - 1) / \gcd(T_r, T_s))$ of phase offsets — so a fixed 32-band sweep misses 15/16 of 3.2 s radars entirely. Randomised search and phase lock remove the trap.
- Detection follows a square-law, non-fluctuating model (Marcum Q): $P_d = 0.9$ needs 11.7 dB SNR at $P_{fa} 10^{-4}$ (13.2 dB at 10^{-6}). Measured P_{fa} in simulation, $0.7\text{--}1.0 \times 10^{-4}$, matches the set value.
- Across 100 held-out missions (20 per scenario suite, none used in training or tuning) ANVESHAK beat the open-loop sweep in 99 of 100 (paired Wilcoxon $p < 10^{-4}$ in every suite). The XGBoost hit model reaches AUC-PR 0.135 against 0.0088 for chance; an externally supplied model scored 0.010. Rules + model beat rules alone on hopping emitters (0.158 vs 0.119, $p < 0.001$).
- Trade-off we report rather than hide: the pop-up radar is found more often (100% vs 95%) but first intercept is later (median 9.0 s vs 3.0 s), because confirm camps delay search; the revisit floor sets the balance.

B. References

1. R. G. Wiley, ELINT: The Interception and Analysis of Radar Signals, Artech House, 2006.
2. J. I. Marcum, “A statistical theory of target detection by pulsed radar,” IRE Trans. Information Theory, 1960.
3. T. Chen, C. Guestrin, “XGBoost: A scalable tree boosting system,” Proc. ACM SIGKDD, 2016.
4. W. R. Thompson, “On the likelihood that one unknown probability exceeds another,” Biometrika, 1933.
5. A. Garivier, E. Moulines, “On upper-confidence bound policies for switching bandit problems,” ALT, 2011.
6. SIH 2026 Problem Statement SIH26055, Defence Research and Development Organisation (DRDO).

Simulation core, hit model and benchmark — Python · NumPy · SciPy · XGBoost, 126 contract tests